

Electric Potential Energy

Goal.

- Understand where electric potential energy comes from: summing over pairs of point charges, or pushing a charge against a field.
- Use that to find the potential energy of an electric dipole in a uniform field.
- See that a system can have *negative* PE, and that “zero” PE is not always the lowest energy available.

1. Potential energy of two point charges (from the class prep)

Pulling a pair of point charges apart, or pushing them together, takes work — work that gets stored as potential energy.

$$U(q_1, q_2) =$$

- As $r_{12} \rightarrow \infty$, $U \rightarrow$ _____.
- For three or more charges, sum over every _____ (not every charge).
- Opposite charges (+ and -): U is _____ at any finite separation, and $U \rightarrow 0$ only as $r \rightarrow \infty$. So $U = 0$ is actually the _____ value here — pulling the charges closer always finds a *lower* (more negative) PE.
- Like charges (same sign): U is _____ at any finite separation, and again $U \rightarrow 0$ as $r \rightarrow \infty$. Here $U = 0$ is the _____ value — you can’t push like charges to a lower PE than infinitely far apart.

$$U_{\text{total}} =$$

Check. Two $+q$ charges are brought from far apart to a separation r . Is U positive or negative? What about a $+q$ and a $-q$?

2. Where is potential energy the lowest?

Draw a ball, and the ground below it.

Check. What is the ball's potential energy? Where is it the lowest?

This is a *uniform* gravitational field — not a hole to fall into. Raising the ball, moving it against the field lines, gives it *more* PE, and that PE is proportional to the distance you raise it.

$$\Delta U_{\text{grav}} =$$

3. The same idea, for a charge in a uniform field

Now replace the ball with a positive charge q , and gravity with a uniform electric field $\vec{E}$. Draw the field, and the charge's starting and ending points, below.

The field pushes the charge with force qE . Push the charge a distance d against the field — the same motion as raising the ball.

$\Delta U =$

- This is exactly like raising the ball: you do _____ work pushing against the field, and that work becomes stored PE, proportional to _____.
- Let the charge go, and it moves *with* the field instead: now ΔU is _____ — the field does the work, and spends the stored PE.

Check. A positive charge is released and moves along $\vec{E}$. Does its PE increase or decrease? Where did the energy go?

4. The dipole in a field

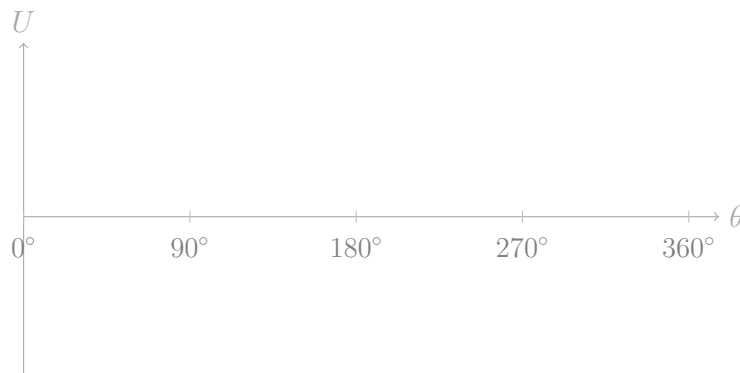
A dipole is a $+q$ and a $-q$ separated by $\vec{d}$ (dipole moment $\vec{p} = q\vec{d}$), sitting in a uniform field $\vec{E}$, tilted at angle θ between $\vec{p}$ and $\vec{E}$. Each charge sits at a different place along $\vec{E}$ — use §3's pushing idea on each charge, then add.

$$U(\theta) =$$

- At $\theta = 90^\circ$ (perpendicular to the field): $U =$ _____.
- At $\theta = 0^\circ$ (aligned with the field): $U =$ _____ — which is _____ than the $\theta = 90^\circ$ value!
- At $\theta = 180^\circ$ (anti-aligned): $U =$ _____.

Check. Which orientation is the most stable (lowest PE)? Is that the orientation where $U = 0$? What does this have in common with the ball in the gravitational field?

5. Sketch it: U vs. θ



Check your curve, and the point-charge and uniform-field results above, against the *Assembling Charges*, *PE vs. Separation*, and *Electric Dipole in a Field* sims — open their **Info** panels.